Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements about physical modeling is most appropriate?

A. A bicycle must always be modeled as a rigid body even when only its center-of-mass path down a long road is needed.
This is a misconception because center-of-mass motion over long distances often does not require rigid-body details.

B. A city bus can be treated as a point particle when estimating how far it travels along a highway over several kilometers.
This is correct because the bus's size is negligible compared with kilometer-scale translation, so a point-mass model captures its position change.

C. If air drag is small, it is never acceptable to approximate the motion as occurring in a vacuum.
This is wrong because "vacuum" is commonly used as an approximation meaning air resistance is neglected.

D. None of the above is correct.
This is incorrect because one of the statements (the point-particle approximation over large distances) is valid.

Q2. Which of the following unit-conversion equations is correct?

A. $1\ \text{mm} = 10^{-2}\ \text{m}$
This mistake confuses millimeters with centimeters (off by a factor of 10).

B. $1\ \text{m} = 0.001\ \text{mm}$
This reverses the conversion direction; meters are larger than millimeters.

C. $1\ \text{mm} = 0.001\ \text{m}$
This is correct because "milli-" means 10^{-3}, so 1 mm = 10^{-3} m.

D. $1000\ \text{mm} = 0.001\ \text{m}$
This is wrong because 1000 mm equals 1 m, not 10^{-3} m.

Q3. Which option lists an incorrect SI unit for the given quantity?

A. Power: watt (W)
This is correct because power is measured in watts (J/s).

B. Momentum: kilogram·meter/second (kg m/s)
This is correct because momentum has dimensions of mass times velocity.

C. Pressure: newton/meter (N/m)
This is incorrect because pressure is force per area, so the SI unit is N/m^2 (pascal), not N/m.

D. Charge: coulomb (C)
This is correct because electric charge is measured in coulombs.

Q4. A scout reports an outpost located at the position vector $\vec{r} = 4\hat{\imath} + 7\hat{\jmath}$, where distances are measured in "megalinks." Which statement is correct?

A. The distance from the origin to the outpost is 8 megalinks.
This is a common rounding/estimation error because the true distance is $\sqrt{65} \approx 8.06$, not exactly 8.

B. The distance from the origin to the outpost is 7 megalinks.
This incorrectly treats one component magnitude as the full distance.

C. The distance from the origin to the outpost is 4 megalinks.
This incorrectly uses only the x-component as the distance from the origin.

D. None of the above is correct.
This is correct because $|\vec{r}| = \sqrt{4^2 + 7^2} = \sqrt{65} \approx 8.06$ megalinks, which is not 8, 7, or 4.

Q5. If the displacement of an object is given by $\vec{s} = \left((0.80\,\mathrm{m/s^3})t^3 - (1.60\,\mathrm{m/s^2})t^2\right)\hat{\imath}$, which statement is correct?

A. This motion has constant acceleration.
This is incorrect because differentiating twice yields an acceleration that depends on t.

B. The magnitude of the acceleration is $1.60\,\mathrm{m/s^2}$ for all t.
This treats a coefficient in $s(t)$ as if it were the (constant) acceleration.

C. The magnitude of the acceleration at $t = 0$ is $0\,\mathrm{m/s^2}$.
This mistake assumes acceleration must vanish at $t = 0$ without actually differentiating.

D. The magnitude of the acceleration at $t = 0$ is $3.20\,\mathrm{m/s^2}$.
This is correct because $a(t) = \frac{d^2s}{dt^2} = (4.80t - 3.20)\,\hat{\imath}$, so at $t = 0$ the magnitude is $3.20\,\mathrm{m/s^2}$.

Q6. A robot moves along a straight line from position $\vec{r}_A = 1.0\hat{\imath} - 2.0\hat{\jmath}$ to $\vec{r}_B = 7.0\hat{\imath} + 5.0\hat{\jmath}$ in 4 s (meters). Which statement is correct?

A. The distance traveled is 9.2 m.
This is correct because the straight-line distance is $|\Delta\vec{r}| = \sqrt{(6)^2 + (7)^2} = \sqrt{85} \approx 9.2\,\mathrm{m}$.

B. The speed at $t = 0$ must be $2.3\,\mathrm{m/s}$.
This is incorrect because knowing only endpoints and total time does not determine the instantaneous speed at $t = 0$.

C. The acceleration at $t = 0$ must be $0.58\,\mathrm{m/s^2}$.
This is incorrect because the motion could have many different acceleration profiles while still connecting the same endpoints in 4 s.

D. None of the above is correct.
This is incorrect because the straight-line distance can be computed uniquely from the given coordinates.

Q7. A soccer ball is kicked with initial speed v_0 at an angle θ above the horizontal (ignore air resistance). What is the ball's speed at its maximum height?

A. v_0
This is incorrect because the vertical component decreases to zero at the top, reducing the speed.

B. $v_0 \cos\theta$
This is correct because at maximum height the vertical velocity is zero, so the speed equals the constant horizontal component $v_0 \cos\theta$.

C. $v_0 \sin\theta$
This is incorrect because $v_0 \sin\theta$ is the initial vertical component, which becomes zero at the top.

D. 0
This is incorrect because only the vertical component is zero at the top; the horizontal component remains.

Q8. A hovercraft starts with initial velocity $\vec{v}_0 = (18.0\,\text{m/s})\hat{\imath}$. Its thrusters then produce a constant acceleration $\vec{a} = (1.50\,\text{m/s}^2)\hat{\imath} + (3.00\,\text{m/s}^2)\hat{\jmath}$. Which statement is correct?

A. The x-component of the hovercraft's velocity remains constant.
This is incorrect because a nonzero a_x changes the x-component of velocity over time.

B. The y-component of the hovercraft's velocity remains constant.
This is incorrect because a nonzero a_y causes the y-component of velocity to change from 0 to nonzero.

C. Both the direction of motion and the speed change.
This is correct because acceleration changes both components of velocity, altering the velocity vector's magnitude and direction.

D. None of the above is correct.
This is incorrect because option C correctly describes the kinematics with two-component acceleration.

Q9. Two point charges exert an electrostatic force on each other. If each charge's magnitude is cut in half and the separation distance is doubled, what happens to the magnitude of the force?

A. It stays the same.
This is incorrect because Coulomb's law scales with both charge and distance changes.

B. It decreases by a factor of 16.
This is correct because $F \propto q_1 q_2 / r^2$ gives a factor $(1/2)(1/2)/(2^2) = 1/16$.

C. It decreases by a factor of 4.
This ignores that the distance change contributes a squared factor in the denominator.

D. It increases by a factor of 4.
This has the scaling direction wrong; reducing charges and increasing distance must reduce the force.

E. It increases by a factor of 16.
This reverses the correct scaling and mistakenly treats the changes as amplifying the force.

Q10. An escape speed from a planet is given by $v = \sqrt{\frac{2GM}{R}}$. Which statement is correct?

A. Doubling the mass of a spacecraft increases the required escape speed.
This is incorrect because the formula depends on the planet's M and R, not on the escaping object's mass.

B. A smaller planet radius always lowers the escape speed.
This is incorrect because decreasing R increases $\frac{GM}{R}$ and thus increases escape speed.

C. A planet with half the mass and half the radius has a higher escape speed.
This is incorrect because M/R is unchanged when both are halved, so the escape speed stays the same.

D. A planet with greater mass and smaller radius has a higher escape speed.
This is correct because v increases when M increases and when R decreases (since $v \propto \sqrt{M/R}$).

Q11. A constant force $\vec{F} = (-2.50\,\text{N})\hat{\imath} + (6.00\,\text{N})\hat{\jmath}$ acts on a particle while it moves along the x-axis from $x = 1.2\,\text{m}$ to $x = 4.8\,\text{m}$. What is the work done by the force?

A. -9.0 J
This is correct because $\Delta\vec{r} = (3.6\,\text{m})\hat{\imath}$ so $W = \vec{F} \cdot \Delta\vec{r} = (-2.50)(3.6) = -9.0\,\text{J}$.

B. 21.6 J
This incorrectly uses the y-component of force even though there is no y-displacement.

C. 9.0 J
This gets the sign wrong by ignoring that F_x is negative while Δx is positive.

D. -21.6 J
This incorrectly multiplies the y-force by the x-displacement as if they were parallel components.

Q12. A 3.0 kg cart moves toward a bumper at 4.0 m/s, hits it, and rebounds at 2.0 m/s in the opposite direction. The contact time is 0.15 s. What is the magnitude of the average force on the cart during the collision?

A. 80 N
This results from using only a partial velocity change (e.g., $4 \to 0$) instead of the full reversal.

B. 120 N
This is correct because $\Delta p = m(v_f - v_i) = 3.0(-2.0 - 4.0) = -18\,\text{N s}$ so $|F_{\text{avg}}| = |\Delta p|/\Delta t = 18/0.15 = 120\,\text{N}$.

C. 40 N
This typically comes from dividing by the wrong time or forgetting that the cart reverses direction.

D. None of the above is correct.
This is incorrect because the impulse-momentum calculation gives a value that is listed.

Q13. A 62 kg person on a frictionless skateboard is initially at rest. They catch a 2.0 kg medicine ball thrown horizontally at 8.0 m/s. After the catch, what is the person's speed (with the ball)?

A. 0.26 m/s
This is correct because momentum conservation for an inelastic catch gives $v_f = \frac{(2.0)(8.0)}{62+2.0} \approx 0.25\,\text{m/s}$ (about $0.26\,\text{m/s}$).

B. 0.13 m/s
This typically comes from mistakenly dividing by only the person's mass and then halving again.

C. 8.0 m/s
This ignores that the catcher is much more massive, so the shared final speed must be far smaller than the ball's initial speed.

D. 4.0 m/s
This incorrectly averages speeds instead of conserving momentum for a perfectly inelastic collision.
